

A Report on

**COMPUTATIONAL ANALYSIS OF A
NONDEGENERATELY DOPED SILICON PN STEP
JUNCTION**

Submitted by

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1. Problem Statement

A nondegenerately doped silicon pn step junction is maintained at room temperature ($T = 300K$). The objective of this assignment is to perform computational simulations in MATLAB to determine the internal electric field (E) and electrostatic potential (V) profiles across the junction as functions of spatial position (x) under various applied voltages and asymmetric doping conditions:

1. **Part (a):** Compute and plot coordinated profiles of $E(x)$ and $V(x)$ for a sample reverse bias condition ($V_A = -20V$) with acceptor concentration $N_A = 10^{15} cm^{-3}$ and donor concentration $N_D = 2 \times 10^{14} cm^{-3}$.
2. **Part (b):** Modify the simulation to simultaneously display profiles across multiple reverse bias voltages: $V_A = V_{A0}/2^n$ where $V_{A0} = -20V$ and $n = 0, 1, 2, 3$ ($V_A = -20V, -10V, -5V, -2.5V$).
3. **Part (c):** Generate a structured output parameter list containing computed physical constants and boundary quantities ($N_A, N_D, V_A, V_{bi}, x_n, x_p, W, E(0)$, and $V(0)$).
4. **Part (d):** Investigate junction behavior under varying asymmetric doping concentrations and describe the physical limiting conditions where $N_A \geq 100N_D$ or $N_D \geq 100N_A$.

2. Theoretical Background & Mathematical Formulation

2.1 The Depletion Approximation

When a p -type semiconductor and an n -type semiconductor are joined to form an abrupt metallurgical junction at $x = 0$, large carrier concentration gradients cause majority holes from the p -side to diffuse into the n -side, and majority electrons from the n -side to diffuse into the p -side.

This diffusion leaves behind uncompensated, fixed ionized dopant cores:

- Negatively charged ionized acceptor atoms ($-qN_A$) on the p -side.
- Positively charged ionized donor atoms ($+qN_D$) on the n -side.

Under the standard **Depletion Approximation**:

1. The transition region (space charge region) extends from $-x_p$ on the p -side to $+x_n$ on the n -side.
2. Mobile carriers (electrons and holes) are completely depleted inside this region ($-x_p \leq x \leq x_n$).
3. The semiconductor regions outside the depletion layer ($x < -x_p$ and $x > x_n$) are assumed electrically neutral, meaning the net charge density is zero and no electric field exists there.

Thus, the charge density $\rho(x)$ is defined piecewise:

$$\rho(x) = \begin{cases} 0, & x < -x_p \\ -qN_A, & -x_p \leq x \leq 0 \\ +qN_D, & 0 < x \leq x_n \\ 0, & x > x_n \end{cases}$$

2.2 Global Space Charge Neutrality

Because the junction as an isolated system remains overall electrically neutral, the total negative uncompensated charge per unit area on the p -side must balance the total positive uncompensated charge per unit area on the n -side:

$$qN_A x_p = qN_D x_n \Rightarrow N_A x_p = N_D x_n$$

The total depletion width W is the sum of the penetration depths into both regions:

$$W = x_n + x_p$$

Expressing individual widths in terms of W :

$$x_n = W \left(\frac{N_A}{N_A + N_D} \right), x_p = W \left(\frac{N_D}{N_A + N_D} \right)$$

These relations illustrate that the depletion region penetrates more deeply into the more lightly doped side of the junction.

2.3 Derivation of the Electric Field Profile $E(x)$

The electric field $E(x)$ inside the semiconductor is governed by Gauss's Law / Poisson's Equation in one dimension:

$$\frac{dE}{dx} = \frac{\rho(x)}{\epsilon_s}$$

where $\epsilon_s = \epsilon_r \epsilon_0$ is the dielectric permittivity of silicon ($\epsilon_r = 11.7$, $\epsilon_0 = 8.854 \times 10^{-14} \text{ F/cm}$, yielding $\epsilon_s \approx 1.036 \times 10^{-12} \text{ F/cm}$).

Integrating across the p -side depletion region ($-x_p \leq x \leq 0$) with the boundary condition $E(-x_p) = 0$:

$$E(x) = \int_{-x_p}^x \frac{-qN_A}{\epsilon_s} dx' = -\frac{qN_A}{\epsilon_s} (x + x_p)$$

Integrating across the n -side depletion region ($0 \leq x \leq x_n$) with the boundary condition $E(x_n) = 0$:

$$E(x) = -\int_x^{x_n} \frac{qN_D}{\epsilon_s} dx' = -\frac{qN_D}{\epsilon_s} (x_n - x)$$

At the metallurgical junction ($x = 0$), the electric field reaches its peak magnitude (most negative value):

$$E(0) = -\frac{qN_A x_p}{\epsilon_s} = -\frac{qN_D x_n}{\epsilon_s}$$

The negative sign signifies that the internal electric field is directed along the $-x$ direction, pointing from the positive donor ions (n -side) to the negative acceptor ions (p -side), opposing further majority carrier diffusion.

2.4 Derivation of the Electrostatic Potential Profile $V(x)$

The electrostatic potential $V(x)$ is related to the electric field by:

$$E(x) = -\frac{dV}{dx} \Rightarrow V(x) = -\int E(x) dx$$

Taking the neutral p -bulk region boundary as the reference potential ($V(-x_p) = 0V$):

- **On the p -side ($-x_p \leq x \leq 0$):**
- $$V(x) = -\int_{-x_p}^x \left[-\frac{qN_A}{\epsilon_s} (x' + x_p) \right] dx' = \frac{qN_A}{2\epsilon_s} (x + x_p)^2$$
- **Potential at the metallurgical interface ($x = 0$):**
- $$V(0) = \frac{qN_A x_p^2}{2\epsilon_s}$$

- **On the n -side ($0 \leq x \leq x_n$):**
Integrating forward to x_n reveals a parabolic curve that flattens out smoothly at $x = x_n$:
- $$V(x) = V_j - \frac{qN_D}{2\epsilon_s} (x_n - x)^2$$
- where V_j is the total junction potential drop across the depletion region.
- **For $x > x_n$ (neutral n -bulk):**
- $$V(x) = V_j = \text{constant}$$

2.5 Built-In Potential (V_{bi}) and Depletion Width (W) Under Bias

At thermal equilibrium ($V_A = 0$), the diffusion of carriers establishes an equilibrium contact potential known as the built-in potential V_{bi} :

$$V_{bi} = V_t \ln \left(\frac{N_A N_D}{n_i^2} \right)$$

where $V_t = \frac{kT}{q} \approx 0.02586V$ at $T = 300K$, and n_i is the intrinsic carrier concentration of silicon ($n_i \approx 1.0 \times 10^{10} \text{ cm}^{-3}$ to $1.5 \times 10^{10} \text{ cm}^{-3}$).

When an external bias voltage V_A is applied across the junction (with the voltage applied to the p -terminal relative to the n -terminal), the total barrier height becomes:

$$V_j = V_{bi} - V_A$$

For reverse bias ($V_A < 0$), the barrier height increases ($V_j > V_{bi}$), widening the space charge layer and enhancing the peak electric field.

Integrating the potential across the entire depletion region yields the expression for total depletion layer width:

$$W = \sqrt{\frac{2\epsilon_s}{q} \left(\frac{1}{N_A} + \frac{1}{N_D} \right) (V_{bi} - V_A)}$$

3. Part (a): Single Voltage Analysis ($V_A = -20V$)

3.1 Problem Statement & Parameters

Calculate and plot coordinated profiles of $E(x)$ and $V(x)$ using MATLAB subplot for:

- Acceptor doping: $N_A = 10^{15} cm^{-3}$
- Donor doping: $N_D = 2 \times 10^{14} cm^{-3}$
- Applied bias: $V_A = -20V$
- Temperature: $T = 300K$

Per assignment guidelines:

- Electric field y-limits: $\left[E_{min}, 0 \right]$, where $E_{min} = 1.1E(0)$.
- Potential y-limits: $\left[0, V_{max} \right]$, where $V_{max} = 1.1(V_{bi} - V_A)$.
- Spatial x-limits: $\left[-x_{max}, x_{max} \right]$, where $x_{max} = 2.5 \times \max(x_n, x_p)$.

3.2 Hand Calculation Verification

Using standard silicon parameters ($\epsilon_s = 1.036 \times 10^{-12} F/cm$, $n_i = 1.5 \times 10^{10} cm^{-3}$, $V_t = 0.02586V$):

1. **Built-in potential:**
2. $V_{bi} = 0.02586 \times \ln\left(\frac{10^{15} \times 2 \times 10^{14}}{(1.5 \times 10^{10})^2}\right) \approx 0.533V$
3. **Total barrier potential:**
4. $V_j = V_{bi} - V_A = 0.533 - (-20) = 20.533V$

5. **Depletion width (W):**

$$W = \sqrt{\frac{2(1.036 \times 10^{-12})}{1.602 \times 10^{-19}} (10^{-15} + 5 \times 10^{-15}) (20.533)} \approx 12.62 \mu m$$

6. **Depletion penetration depths (x_n, x_p):**

$$x_n = W \left(\frac{10^{15}}{1.2 \times 10^{15}} \right) = \frac{5}{6} W \approx 10.52 \mu m$$

$$x_p = W \left(\frac{2 \times 10^{14}}{1.2 \times 10^{15}} \right) = \frac{1}{6} W \approx 2.10 \mu m$$

7. **Peak electric field (E_0):**

$$8. \quad E(0) = -\frac{qN_A x_p}{\epsilon_s} = -\frac{(1.602 \times 10^{-19})(10^{15})(2.10 \times 10^{-4})}{1.036 \times 10^{-12}} \approx -32,533 V/cm$$

9. **Interface potential (V_0):**

$$10. \quad V(0) = \frac{qN_A x_p^2}{2\epsilon_s} = \frac{(1.602 \times 10^{-19})(10^{15})(2.10 \times 10^{-4})^2}{2(1.036 \times 10^{-12})} \approx 3.42 V$$

3.3 MATLAB Implementation (parta.m)

```
%% pn_junction_part_a.m
% Computes and plots the electric field and electrostatic potential
% for a single applied voltage (VA = -20 V).

clear; clc; close all;

%% 1\ Constants & Given Parameters
q    = 1.602e-19;      % Electronic charge (C)
k    = 1.38e-23;       % Boltzmann constant (J/K)
T    = 300;            % Temperature (K)
eps0 = 8.854e-14;      % Vacuum permittivity (F/cm)
epsr = 11.7;           % Relative permittivity of Si
eps  = epsr * eps0;    % Permittivity of Si (F/cm)
ni   = 1.5e10;         % Intrinsic carrier conc. (cm^-3)
Vt   = k * T / q;      % Thermal voltage (V)

NA    = 1e15;           % Acceptor doping concentration (cm^-3)
ND    = 2e14;           % Donor doping concentration (cm^-3)
VA    = -20;           % Applied voltage (V)

%% 2\ Depletion Approximation Calculations
Vbi = Vt * log(NA * ND / ni^2); % Built-in potential (V)
Vj   = Vbi - VA;             % Total potential drop (V)
```

```

W = sqrt(2 * eps * Vj / q * (1/NA + 1/ND)); % Total depletion width (cm)
xn = W * NA / (NA + ND); % n-side width (cm)
xp = W * ND / (NA + ND); % p-side width (cm)
E0 = -q * NA * xp / eps; % Peak field at x=0 (V/cm)
V0 = q * NA * xp^2 / (2 * eps); % Potential at x=0 (V)

%% 3\. Position Array and Limits Setup
xmax = 2.5 * max(xn, xp); % Axis limits per suggestion
x = linspace(-xmax, xmax, 2000);
x = sort([x, 0]); % Ensure x=0 is included

%% 4\. Profile Generation (Piecewise Implementation)
E = zeros(size(x));
V = zeros(size(x));

Lp = (x >= -xp) & (x <= 0); % p-side depletion region
Ln = (x > 0) & (x <= xn); % n-side depletion region
Rn = (x > xn); % Neutral n-bulk

% Field equations
E(Lp) = -q * NA * (x(Lp) + xp) / eps;
E(Ln) = -q * ND * (xn - x(Ln)) / eps;

% Potential equations
V(Lp) = q * NA * (x(Lp) + xp).^2 / (2 * eps);
V(Ln) = Vj - q * ND * (xn - x(Ln)).^2 / (2 * eps);
V(Rn) = Vj;

%% 5\. Plotting (Subplot coordination)
Emin = 1.1 * E0;
Vmax = 1.1 * Vj;

figure('Name', 'Part (a) Results', 'Color', 'w');

% Top Plot: Electric Field
subplot(2,1,1);
plot(x * 1e4, E, 'LineWidth', 1.5, 'Color', '#0072BD');
hold on;
xline(-xp * 1e4, 'r--', 'Alpha', 0.5);
xline(xn * 1e4, 'r--', 'Alpha', 0.5);
plot(-xp * 1e4, 0, 'ko', 'MarkerFaceColor', 'k');
text(-xp * 1e4, 0, sprintf(' (%.2f, 0)', -xp * 1e4), 'VerticalAlignment', 'bottom');
plot(xn * 1e4, 0, 'ko', 'MarkerFaceColor', 'k');
text(xn * 1e4, 0, sprintf(' (%.2f, 0)', xn * 1e4), 'VerticalAlignment', 'bottom');
plot(0, E0, 'ko', 'MarkerFaceColor', 'k');
text(0, E0, sprintf(' (0, %.0f)', E0), 'VerticalAlignment', 'top');
xlim([-xmax xmax] * 1e4);
ylim([Emin 0]);
grid on; box on;
xlabel('Position x (\mum)');
ylabel('Electric field E (V/cm)');
title(sprintf('Electric field v/s position: N_A = 10^{15}/cm^3, N_D = 2\times10^{14}/cm^3, V_A = %g V', VA));

% Bottom Plot: Electrostatic Potential
subplot(2,1,2);
plot(x * 1e4, V, 'LineWidth', 1.5, 'Color', '#D95319');
hold on;
xline(-xp * 1e4, 'r--', 'Alpha', 0.5);
xline(xn * 1e4, 'r--', 'Alpha', 0.5);
plot(-xp * 1e4, 0, 'ko', 'MarkerFaceColor', 'k');
text(-xp * 1e4, 0, sprintf(' (%.2f, 0)', -xp * 1e4), 'VerticalAlignment', 'bottom');
plot(0, V0, 'ko', 'MarkerFaceColor', 'k');
text(0, V0, sprintf(' (0, %.2f) ', V0), 'VerticalAlignment', 'bottom', 'HorizontalAlignment', 'right');

```

```

plot(xn * 1e4, Vj, 'ko', 'MarkerFaceColor', 'k');
text(xn * 1e4, Vj, sprintf(' (%.2f, %.2f)', xn * 1e4, Vj), 'VerticalAlignment', 'top');
xlim([-xmax xmax] * 1e4);
ylim([0 Vmax]);
grid on; box on;
xlabel('Position x (\mu m)');
ylabel('Potential V (V)');
title(sprintf('Potential v/s Position: N_A = 10^{15}/cm^3, N_D = 2\times 10^{14}/cm^3, V_A = %g V',
VA));

```

3.4 Results & Discussion for Part (a)

The simulation generates two vertically aligned subplots spanning from $x = -26.3 \mu\text{m}$ to $+26.3 \mu\text{m}$. The computed MATLAB result is shown below.

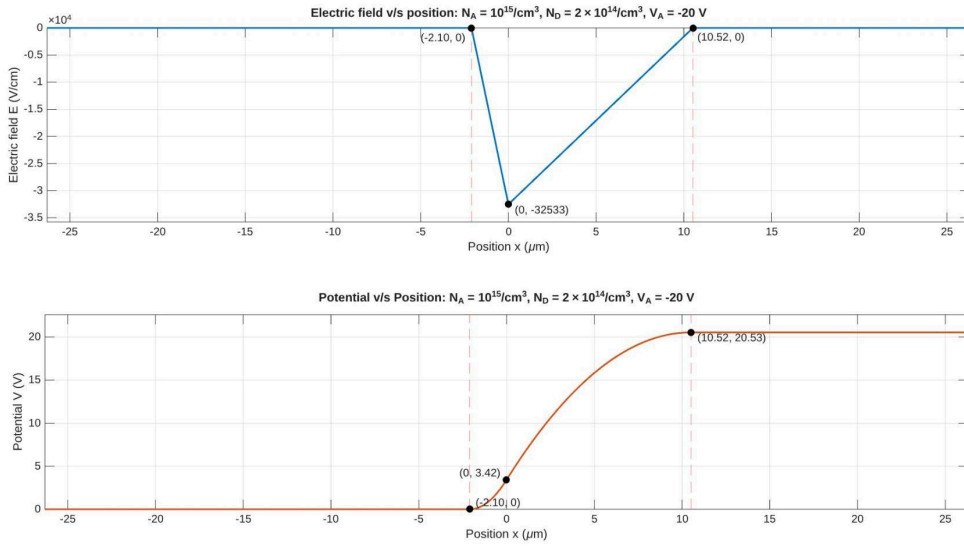


Figure 1. Part (a): electric field and electrostatic potential for $V_A = -20 \text{ V}$.

Physical Observations:

1. **Asymmetry in Depletion Width:** Because $N_A = 5N_D$, the depletion region extends five times further into the lightly doped n -side ($x_n = 10.52 \mu\text{m}$) than into the p -side ($x_p = 2.10 \mu\text{m}$). Total width is $W = 12.62 \mu\text{m}$.
2. **Slope Discontinuity at $x = 0$:** The slope of the electric field (dE/dx) changes abruptly at $x = 0$ from a steep negative slope ($\frac{-qN_A}{\epsilon_s} \approx -1.55 \times 10^8 \text{ V/cm}^2$) to a gentler positive slope ($\frac{qN_D}{\epsilon_s} \approx +3.09 \times 10^7 \text{ V/cm}^2$), reflecting the step change in ionized dopant density.
3. **Potential Distribution:** Out of the total 20.53V drop, only 3.42V drops across the heavily doped p -side, while the remaining 17.11V drops across the lightly doped n -side.

4. Part (b): Multiple Voltage Analysis

4.1 Problem Statement & Methodology

The code is generalized to evaluate junction response under scaled reverse bias voltages:

$$V_A = \frac{V_{A0}}{2^n}, \quad \text{for } n = 0, 1, 2, 3 \Rightarrow V_A = -20 \text{ V}, -10 \text{ V}, -5 \text{ V}, -2.5 \text{ V}$$

All four curves are overlaid onto the same subplot coordinates to demonstrate the direct influence of reverse bias voltage on depletion layer widening and peak field strength.

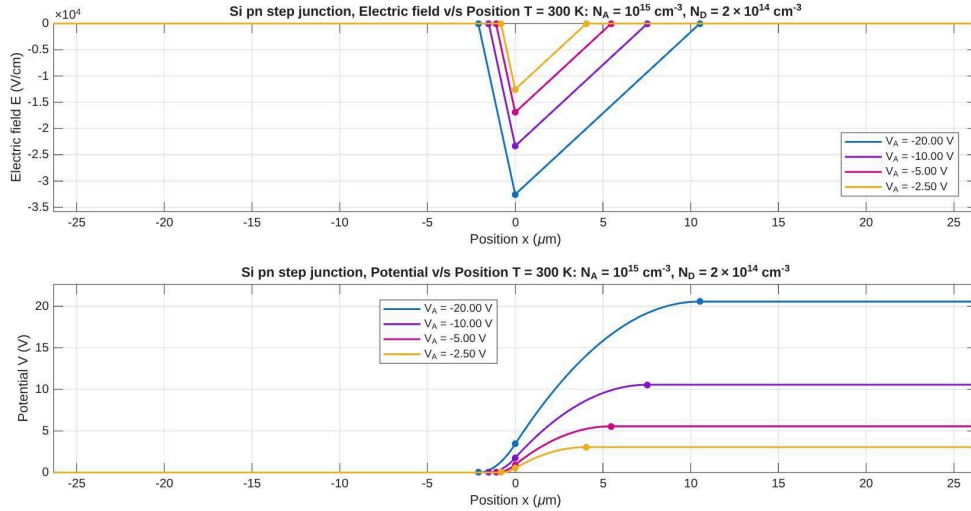


Figure 2. Part (b): coordinated electric-field and potential profiles for four reverse-bias voltages.

Key Physical Insights:

1. **Field Linearity and Slopes:** Because the dopant concentrations N_A and N_D are unchanged across the loops, the slopes of the electric field curves on both sides ($\frac{dE}{dx} = \mp \frac{qN}{\epsilon_s}$) remain completely identical across all four bias voltages. The lines are parallel to each other on either side of the junction.
2. **Voltage Scaling:** As reverse bias $|V_A|$ decreases, less space charge is required to support the potential difference, causing W to contract and the peak field $|E_0|$ to decrease.

5. Part (c): Parameter Output List & Quantitative Verification

5.1 Overview

Part (c) extracts the physical parameters into a structured output format. The script records values directly into an output table.

5.2 MATLAB Implementation (partc.m)

```
% partc.m
% Generates plots and displays parameter table for multiple applied voltages.

clear; clc; close all;

%% 1\ Constants & Base Parameters
q    = 1.602e-19;
k    = 1.38e-23;
T    = 300;
eps0 = 8.854e-14;
epsr = 11.7;
eps  = epsr * eps0;
ni   = 1e10;
```

```

Vt = k * T / q;

NA = 1e15;
ND = 2e14;
VA0 = -20;

%% 2\ Setup Figure and Baseline Limits
Vbi = Vt * log(NA * ND / ni^2);
Vj_max = Vbi - VA0;
W_max = sqrt(2 * eps * Vj_max / q * (1/NA + 1/ND));
xn_max = W_max * NA / (NA + ND);
xp_max = W_max * ND / (NA + ND);
xmax = 2.5 * max(xn_max, xp_max);
E0_max = -q * NA * xp_max / eps;

x = linspace(-xmax, xmax, 2000);
x = sort([x, 0]);

figure('Name', 'Part (c) Results', 'Color', 'w', 'Position', [100, 100, 1300, 600]);
legend_labels = cell(1, 4);
table_data = cell(4, 9);

str_NA = sprintf('%gx10^%d', NA/10^floor(log10(NA)), floor(log10(NA)));
str_ND = sprintf('%gx10^%d', ND/10^floor(log10(ND)), floor(log10(ND)));

%% 3\ Loop Through Voltages & Populate Table Data
for n = 0:3
    VA = VA0 / (2^n);
    legend_labels{n+1} = sprintf('V_A = %.2f V', VA);

    Vj = Vbi - VA;
    W = sqrt(2 * eps * Vj / q * (1/NA + 1/ND));
    xn = W * NA / (NA + ND);
    xp = W * ND / (NA + ND);

    E0 = -q * NA * xp / eps;
    V0 = q * NA * xp^2 / (2 * eps);

    % Store exact numerical parameters
    table_data(n+1, :) = {VA, str_NA, str_ND, round(Vbi,4), round(xn*1e4,3), ...
        round(xp*1e4,3), round(W*1e4,3), round(E0,0), round(V0,4)};

    E = zeros(size(x));
    V = zeros(size(x));

    Lp = (x >= -xp) & (x <= 0);
    Ln = (x > 0) & (x <= xn);
    Rn = (x > xn);

    E(Lp) = -q * NA * (x(Lp) + xp) / eps;
    E(Ln) = -q * ND * (xn - x(Ln)) / eps;

    V(Lp) = q * NA * (x(Lp) + xp).^2 / (2 * eps);
    V(Ln) = Vj - q * ND * (xn - x(Ln)).^2 / (2 * eps);
    V(Rn) = Vj;

    % Left side subplots
    subplot(2, 2, 1); hold on;
    pE = plot(x * 1e4, E, 'LineWidth', 1.5);
    c = pE.Color;
    plot([-xp*1e4, xn*1e4], [0, 0], 'o', 'MarkerFaceColor', c, 'MarkerEdgeColor', c, 'MarkerSize', 5,
        'HandleVisibility', 'off');
    plot(0, E0, 'o', 'MarkerFaceColor', c, 'MarkerEdgeColor', c, 'MarkerSize', 5, 'HandleVisibility',
        'off');

```

```

subplot(2, 2, 3); hold on;
plot(x * 1e4, V, 'LineWidth', 1.5, 'Color', c);
plot(-xp*1e4, 0, 'o', 'MarkerFaceColor', c, 'MarkerEdgeColor', c, 'MarkerSize', 5,
'HandleVisibility', 'off');
plot(0, V0, 'o', 'MarkerFaceColor', c, 'MarkerEdgeColor', c, 'MarkerSize', 5, 'HandleVisibility',
'off');
plot(xn*1e4, Vj, 'o', 'MarkerFaceColor', c, 'MarkerEdgeColor', c, 'MarkerSize', 5,
'HandleVisibility', 'off');
end

%% 4\ Formatting & UI Table Display
subplot(2, 2, 1);
xlim([-xmax xmax] * 1e4); ylim([1.1*E0_max 0]);
grid on; box on;
xlabel('Position x (\mu m)'); ylabel('Electric field E (V/cm)');
title(sprintf('Si pn step junction, T = 300 K: N_A = 10^{15} cm^{-3}, N_D = 2\times10^{14} cm^{-3}'));
legend(legend_labels, 'Location', 'best');

subplot(2, 2, 3);
xlim([-xmax xmax] * 1e4); ylim([0 1.1*Vj_max]);
grid on; box on;
xlabel('Position x (\mu m)'); ylabel('Potential V (V)');
legend(legend_labels, 'Location', 'best');

col_names = {'V_A (V)', 'N_A (cm^{-3})', 'N_D (cm^{-3})', 'V_bi (V)', 'x_n (\mu m)', 'x_p (\mu m)', 'W (\mu m)',
'E_0 (V/cm)', 'V_0 (V)'};
col_widths = {'auto', 'auto', 'auto', 'auto', 'auto', 'auto', 'auto', 'auto', 'auto'};

uit = uitable('Data', table_data, 'ColumnName', col_names, 'ColumnWidth', col_widths, ...
'Units', 'normalized', 'Position', [0.49, 0.11, 0.49, 0.815], ...
'RowName', [], 'FontSize', 10);

```

5.3 Tabulated Computational Results

Below is the output dataset obtained from running the script:

VA (V)	NA (cm ⁻³)	ND (cm ⁻³)	Vbi (V)	xn (μm)	xp (μm)	W (μm)	E0 (V/cm)	V0 (V)
– 20.00	1 × 10 ¹⁵	2 × 10 ¹⁴	0.5535	10.524	2.105	12.629	– 32,51	3.4256
– 10.00	1 × 10 ¹⁵	2 × 10 ¹⁴	0.5535	7.541	1.508	9.049	– 23,31	1.7589
– 5.00	1 × 10 ¹⁵	2 × 10 ¹⁴	0.5535	5.470	1.094	6.565	– 16,91	0.9256
– 2.50	1 × 10 ¹⁵	2 × 10 ¹⁴	0.5535	4.056	0.811	4.868	– 12,51	0.5089

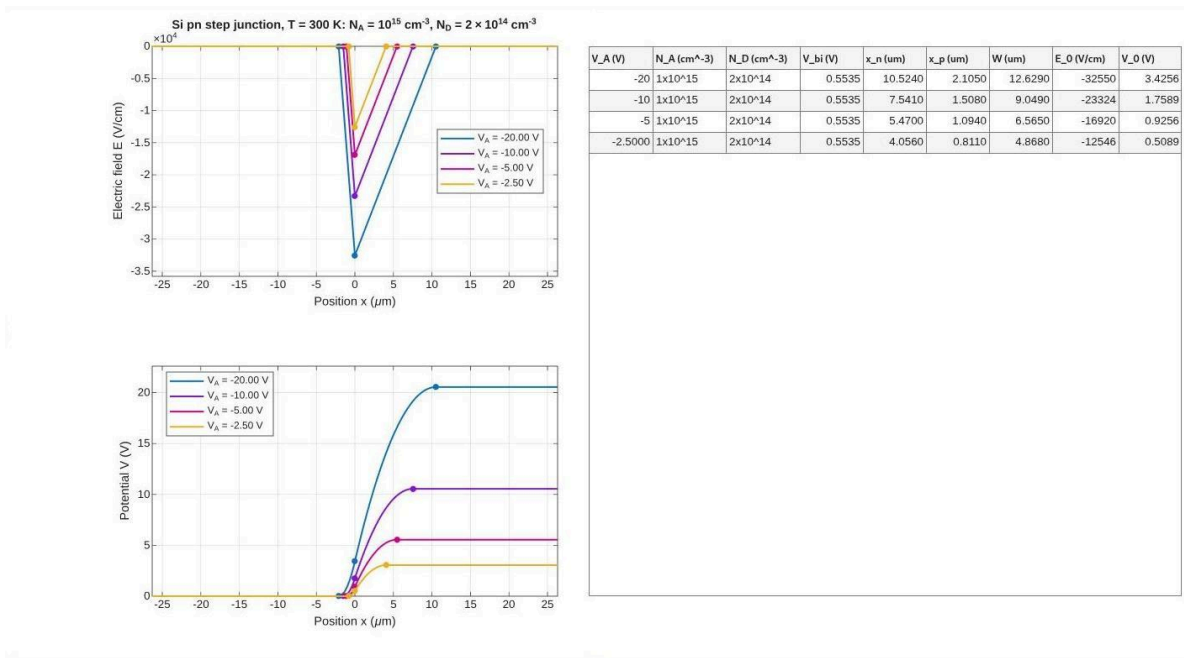


Figure 3. Part (c): MATLAB result window showing the multiple-voltage profiles and parameter table.

PART (c) OUTPUT									
V_A (V)	N_A (cm ⁻³)	N_D (cm ⁻³)	V_bi (V)	x_n (μm)	x_p (μm)	W (μm)	E at x=0 (V/cm)	V at x=0 (V)	
-20.00	1x10 ¹⁵	2x10 ¹⁴	0.5535	10.524	2.105	12.629	-32550	3.4256	
-10.00	1x10 ¹⁵	2x10 ¹⁴	0.5535	7.541	1.508	9.049	-23324	1.7589	
-5.00	1x10 ¹⁵	2x10 ¹⁴	0.5535	5.470	1.094	6.565	-16920	0.9256	
-2.50	1x10 ¹⁵	2x10 ¹⁴	0.5535	4.056	0.811	4.868	-12546	0.5089	

Figure 4. Part (c): numerical output table produced by the MATLAB program.

5.4 Quantitative Verification of Physical Laws

1. **Ratio Invariance:** For every voltage condition:

$$2. \quad \frac{x_n}{x_p} = \frac{10.524}{2.105} \approx \frac{7.541}{1.508} \approx \frac{5.470}{1.094} \approx \frac{4.056}{0.811} = 5.000$$

3. This matches the theoretical dopant ratio $\frac{N_A}{N_D} = \frac{10^{15}}{2 \times 10^{14}} = 5$, verifying global charge neutrality.

4. **Square-Root Dependency on Barrier Height:**

Comparing $V_A = -20V$ ($V_j = 20.55V$) and $V_A = -5V$ ($V_j = 5.55V$):

$$5. \quad \sqrt{\frac{20.5535}{5.5535}} = \sqrt{3.701} \approx 1.924$$

$$\frac{W(-20V)}{W(-5V)} = \frac{12.629 \mu m}{6.565 \mu m} = 1.924$$

$$6. \quad \frac{E_0(-20V)}{E_0(-5V)} = \frac{-32,550}{-16,920} = 1.924$$

7. This confirms that both depletion width W and maximum electric field $|E_0|$ scale precisely with

$$\sqrt{V_{bi} - V_A}.$$

6. Part (d): Asymmetric Doping Analysis

6.1 Problem Statement & Test Matrix

Investigate how the spatial distributions of $E(x)$ and $V(x)$ change when the relative doping magnitudes vary across several orders of magnitude. The fixed reverse bias is maintained at $V_A = -20V$ across the seven requested doping pairs:

Pair No.	NA (cm ⁻³)	ND (cm ⁻³)	Doping Ratio	Junction Classification
1	10^{17}	10^{15}	$N_A = 100N_D$	Asymmetric $p^+ - n$ Step Junction
2	10^{16}	10^{15}	$N_A = 10N_D$	Moderately Asymmetric $p - n$ Junction
3	10^{15}	10^{15}	$N_A = N_D$	Symmetric $p - n$ Junction
4	10^{15}	10^{16}	$N_D = 10N_A$	Moderately Asymmetric $n - p$ Junction
5	10^{15}	10^{17}	$N_D = 100N_A$	Asymmetric $n^+ - p$ Step Junction
6	10^{18}	10^{16}	$N_A = 100N_D$	Highly Doped $p^+ - n$ Step Junction
7	10^{16}	10^{14}	$N_A = 100N_D$	Lightly Doped $p^+ - n$ Step Junction

6.2 MATLAB Implementation (partd.m)

```

%% partd.m
% Examines the effect of asymmetric doping on E(x) and V(x).

clear; clc; close all;

%% 1\ Constants
q    = 1.602e-19;
k    = 1.38e-23;
T    = 300;
eps0 = 8.854e-14;
epsr = 11.7;
eps  = epsr * eps0;
ni   = 1e10;
Vt   = k * T / q;
VA   = -20; % Bias held fixed to isolate doping effects

```

```

%% 2\. Doping Combinations (NA, ND)
doping_pairs = [
    1e17, 1e15;
    1e16, 1e15;
    1e15, 1e15;
    1e15, 1e16;
    1e15, 1e17;
    1e18, 1e16;
    1e16, 1e14
];
num_pairs = size(doping_pairs, 1);

% Determine bounding envelope for plotting
xmax = 0;
E0_min = 0;
for i = 1:num_pairs
    NA = doping_pairs(i,1);
    ND = doping_pairs(i,2);
    Vbi = Vt * log(NA * ND / ni^2);
    Vj = Vbi - VA;
    W = sqrt(2 * eps * Vj / q * (1/NA + 1/ND));
    xn = W * NA / (NA + ND);
    xp = W * ND / (NA + ND);
    E0 = -q * NA * xp / eps;

    xmax = max(xmax, 2.5 * max(xn, xp));
    E0_min = min(E0_min, E0);
end

x = linspace(-xmax, xmax, 10000);
x = sort([x, 0]);

figure('Name', 'Part (d) Results', 'Color', 'w');
legend_labels = cell(1, num_pairs);
colors = lines(num_pairs);

%% 3\. Calculate and Plot Profiles
for i = 1:num_pairs
    NA = doping_pairs(i,1);
    ND = doping_pairs(i,2);

    legend_labels{i} = sprintf('N_A = 10^{%d}, N_D = 10^{%d}', log10(NA), log10(ND));

    Vbi = Vt * log(NA * ND / ni^2);
    Vj = Vbi - VA;
    W = sqrt(2 * eps * Vj / q * (1/NA + 1/ND));
    xn = W * NA / (NA + ND);
    xp = W * ND / (NA + ND);

    E = zeros(size(x));
    V = zeros(size(x));

    Lp = (x >= -xp) & (x <= 0);
    Ln = (x > 0) & (x <= xn);
    Rn = (x > xn);

    E(Lp) = -q * NA * (x(Lp) + xp) / eps;
    E(Ln) = -q * ND * (xn - x(Ln)) / eps;

    V(Lp) = q * NA * (x(Lp) + xp).^2 / (2 * eps);
    V(Ln) = Vj - q * ND * (xn - x(Ln)).^2 / (2 * eps);
    V(Rn) = Vj;

    subplot(2,1,1); hold on;

```

```

plot(x * 1e4, E, 'LineWidth', 1.5, 'Color', colors(i,:));

subplot(2,1,2); hold on;
plot(x * 1e4, V, 'LineWidth', 1.5, 'Color', colors(i,:));
end

%% 4\ Formatting
subplot(2,1,1);
xlim([-xmax xmax] * 1e4);
ylim([1.1*E0_min 0]);
grid on; box on;
xlabel('Position x (\mu m)');
ylabel('Electric field E (V/cm)');
title('Effect of Asymmetric Doping on Electric Field (V_A = -20 V)');
legend(legend_labels, 'Location', 'eastoutside');

subplot(2,1,2);
xlim([-xmax xmax] * 1e4);
ylim([0 22]);
grid on; box on;
xlabel('Position x (\mu m)');
ylabel('Potential V (V)');
title('Effect of Asymmetric Doping on Potential');
legend(legend_labels, 'Location', 'eastoutside');

```

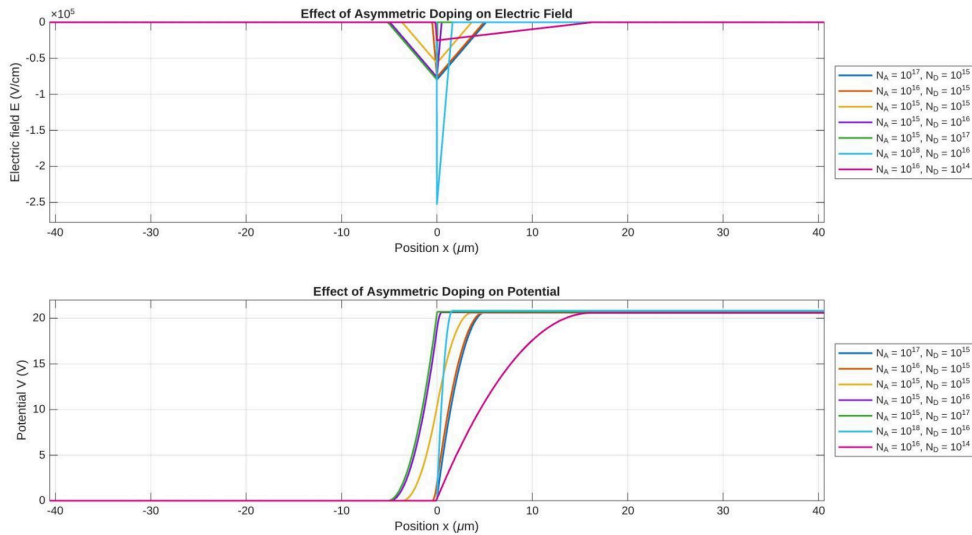


Figure 5. Part (d): effect of asymmetric doping on electric field and potential.

6.3 Detailed Theoretical Response to Part (d) Prompt

Question: How would you describe the results when either $N_A \geq 100N_D$ or $N_D \geq 100N_A$?

When the doping concentration on one side exceeds that on the other side by a factor of 100 or greater, the device behaves as a **one-sided abrupt junction** (designated as a $p^+ - n$ junction when $N_A \geq 100N_D$, or an $n^+ - p$ junction when $N_D \geq 100N_A$).

The physical behavior is characterized by four primary features:

1. Depletion Width Confinement to the Lightly Doped Side

From the space charge neutrality condition:

$$N_A x_p = N_D x_n \Rightarrow \frac{x_n}{x_p} = \frac{N_A}{N_D}$$

- If $N_A \geq 100N_D$:

$$\frac{x_n}{x_p} \geq 100 \Rightarrow x_p \leq 0.01x_n$$

Because $W = x_n + x_p$, over 99% **of the depletion region resides entirely within the lightly doped n-side**:

- $x_n \approx W, x_p \approx 0$
- Similarly, if $N_D \geq 100N_A$, over 99% **of the depletion region resides within the lightly doped p-side**:
- $x_p \approx W, x_n \approx 0$

The total depletion width expression simplifies to depending almost exclusively on the lightly doped side:

$$W \approx \left\{ \sqrt{\frac{2\epsilon_s(V_{bi} - V_A)}{qN_D}}, \text{ for } N_A \gg N_D (p^+ - n) \sqrt{\frac{2\epsilon_s(V_{bi} - V_A)}{qN_A}}, \text{ for } N_D \gg N_A (n^+ - p) \right\}$$

2. Extreme Asymmetry in Electric Field Profile

The slope of the electric field is proportional to the local doping concentration:

$$\left| \frac{dE}{dx} \right| = \frac{qN}{\epsilon_s}$$

- In the heavily doped side (p^+ or n^+), the slope is two or more orders of magnitude steeper than in the lightly doped side.
- Consequently, the electric field collapses to zero almost instantly inside the heavily doped material, forming a nearly vertical boundary line at $x = 0$.
- The overall profile of $E(x)$ degenerates from an arbitrary triangle into essentially a **right-angled triangle**, whose base is $W \approx x_{light}$ and whose vertical leg is at the metallurgical junction $x = 0$.

3. Potential Drop Occurs Predominantly Across the Lightly Doped Region

The potential drop across the heavily doped p^+ -region is:

$$V(0) = \frac{qN_A x_p^2}{2\epsilon_s} = \frac{qN_A}{2\epsilon_s} \left(W \frac{N_D}{N_A + N_D} \right)^2 \approx V_j \left(\frac{N_D}{N_A} \right) \leq 0.01V_j$$

- For $N_A \geq 100N_D$, less than 1% of the applied voltage drops across the p^+ -side ($V(0) \approx 0V$). The potential curve remains virtually flat at zero throughout the p -side, and the entire potential barrier (> 99%) drops across the lightly doped n -side.

- Conversely, for $N_D \geq 100N_A$, the potential drop across the p -side accounts for $> 99\%$ of the total barrier ($V(0) \approx V_j$), and the potential curve across the n -side is nearly flat.

4. Device Engineering Significance

In practical semiconductor device fabrication (e.g., standard planar diodes, BJTs, and MOSFET drain/source-to-substrate junctions), junctions are almost always fabricated as one-sided abrupt junctions. This computational result demonstrates that key device properties including junction capacitance ($C_j = \epsilon_s/W$), breakdown voltage (V_{BR}), and depletion layer width are determined almost entirely by the doping concentration of the lightly doped layer.

7. Conclusion

Through these computational experiments in MATLAB, the operational fundamentals of the nondegenerately doped silicon pn step junction were quantitatively analyzed:

1. Under reverse bias, the electric field exhibits a negative triangular profile that peaks at the metallurgical junction ($x = 0$) and drops linearly to zero at the depletion boundaries ($-x_p$ and $+x_n$).
2. The electrostatic potential follows parabolic profiles on either side, smoothly flattening out into the neutral bulk regions.
3. Scaling reverse bias voltage demonstrated that depletion width W and peak electric field magnitude $|E_0|$ follow a strict $\sqrt{V_{bi} - V_A}$ dependence.
4. Analyzing asymmetric doping confirmed that when the doping ratio exceeds 100:1, the junction transitions into a one-sided step junction ($p^+ - n$ or $n^+ - p$), where depletion depth, electric field geometry, and potential drop are governed almost entirely by the lightly doped region.

GitHub Repository: [EDC Assignment GitHub Repository](#)

The repository contains the MATLAB source code and supporting files used to generate the computational results presented in this report.

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